INTERN NAME : Nallamilli Rajasri

INTERN ID : 03

FRONTEND LIVE URL : https://pillsync-git-main-nallamilli-rajasri-s-projects.vercel.app/

PROBLEM STATEMENT:

Managing multiple medications, remembering dosage schedules, and tracking medicine intake can be challenging. Missed doses and delayed refills may affect treatment consistency. Caregivers also need an efficient way to monitor patients and manage multiple profiles.

PillSync provides a centralized platform for medication management, reminders, adherence tracking, refill prediction, and caregiver monitoring, helping users manage medications efficiently and securely.

TECH STACK :

A. FRONTEND TECHNOLOGIES

Technology	Purpose in PillSync
React 19	Building the user interface and reusable components
Vite	Frontend development server and production build tool
Tailwind CSS 4	Styling the application and creating layouts
Redux Toolkit	Managing application state
React Redux	Connecting React components to the Redux store
React Router DOM	Navigating between application pages
Lucide React	Providing icons for the interface
JavaScript (ES Modules)	Implementing frontend functionality
HTML and CSS	Structuring and styling the user interface
Browser Local Storage	Storing certain medication, profile, and history data locally

B. BACKEND TECHNOLOGIES

Technology	Purpose in PillSync
Python	Backend application programming
Django 5.0.10	Backend framework and application structure
Django REST Framework	Developing REST APIs and API endpoints

Django REST Framework SimpleJWT	JWT-based authentication and token management
Django Sessions	Supporting session-based authentication
Django CORS Headers	Configuring cross-origin access
Django ORM	Interacting with the application's database
Python-dotenv	Loading environment-based configuration
PostgreSQL driver (psycopg)	Supporting PostgreSQL database connections

C. DATABASE TECHNOLOGIES

Technology	Purpose
PostgreSQL	Configured database option for persistent application data
SQLite	Local development fallback
Django ORM	Database queries and model operations
Django Migrations	Managing database schema changes

D. AUTHENTICATION & SECURITY

Technology	Purpose
JWT	Access tokens and authenticated API requests
SimpleJWT Token Blacklisting	Supporting refresh-token invalidation during logout
OAuth/Google authentication workflow	Providing an endpoint for Google ID-token exchange
Django Authentication	User authentication and session support
Custom DRF Permissions	Controlling access based on user roles and patient-profile relationships

E. DEVELOPMENT & DEPLOYMENT TOOLS

Technology	Purpose
Git and GitHub	Source-code management and collaboration
Docker	Containerizing application components
Docker Compose	Configuring multi-container development or deployment
GitHub Actions	Automating repository checks and CI workflows
Postman	API testing
ESLint	Frontend code-quality checks
Pytest	Backend testing framework

FUNCTIONALITIES I HAVE DEVELOPED :

1. Frontend Application Setup

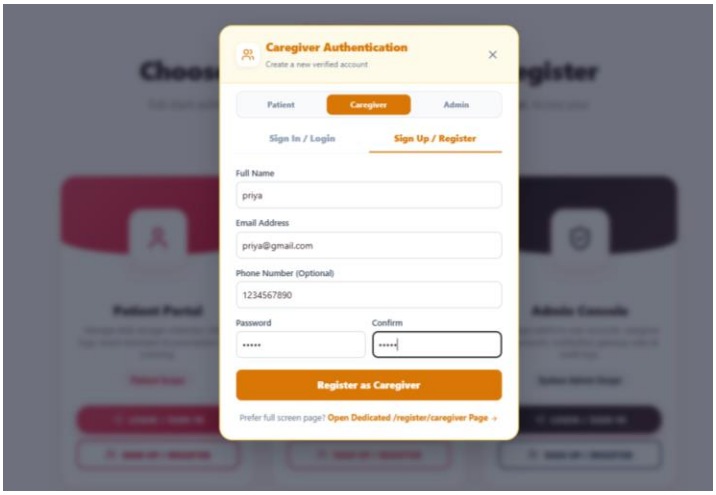
Functionalities include:

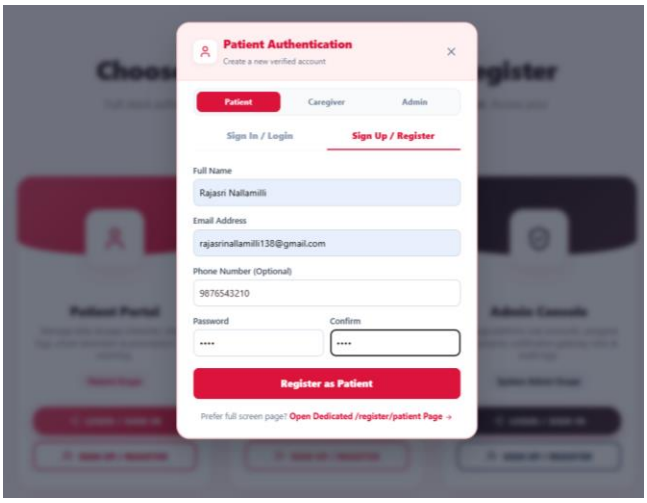
- React application initialization using Vite.
- Tailwind CSS integration for styling.
- Redux Toolkit configuration for state management.
- React Router configuration for navigation.
- Separate page structures for patient, caregiver, and admin portals.
- Reusable components and feature-specific service files.
- A frontend production-build command.

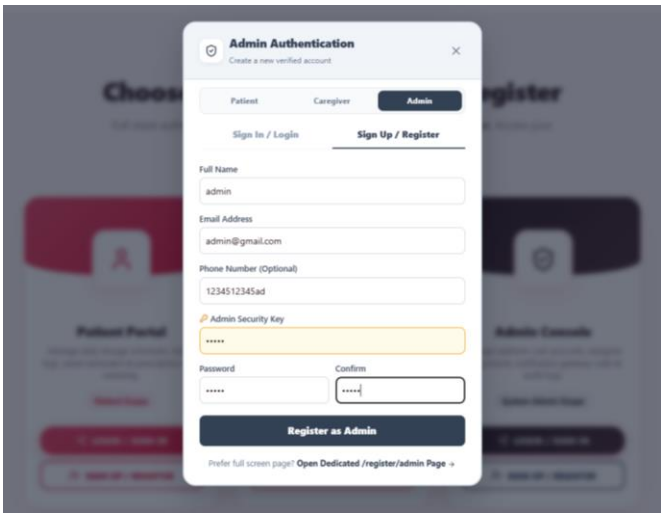
2. Authentication and User Management

Functionalities include:

- User registration.
- User login.
- JWT access-token authentication.
- Refresh-token handling.
- Logout with refresh-token blacklisting.
- Google ID-token exchange endpoint.
- Password change.
- Password reset request and confirmation endpoints.
- Current-user retrieval.
- Separate login and registration pages for patients, caregivers, and administrators.







3. Role-Based Access Control

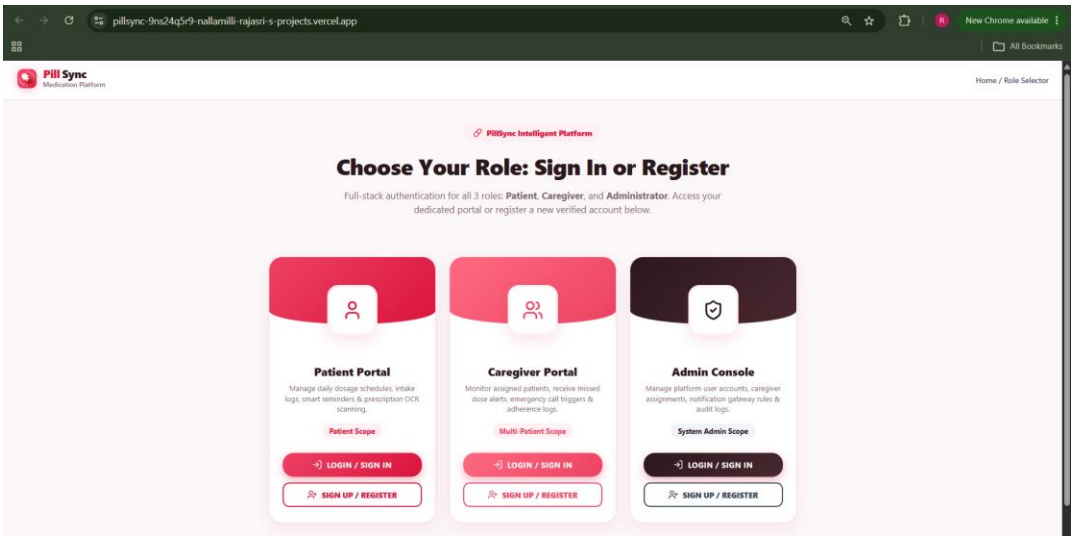
I have defined three principal user roles:

- Patient
- Caregiver
- Admin

Functionalities include:

- Checking whether a user has a required role.
- Defining patient-specific permissions.
- Defining caregiver-specific permissions.

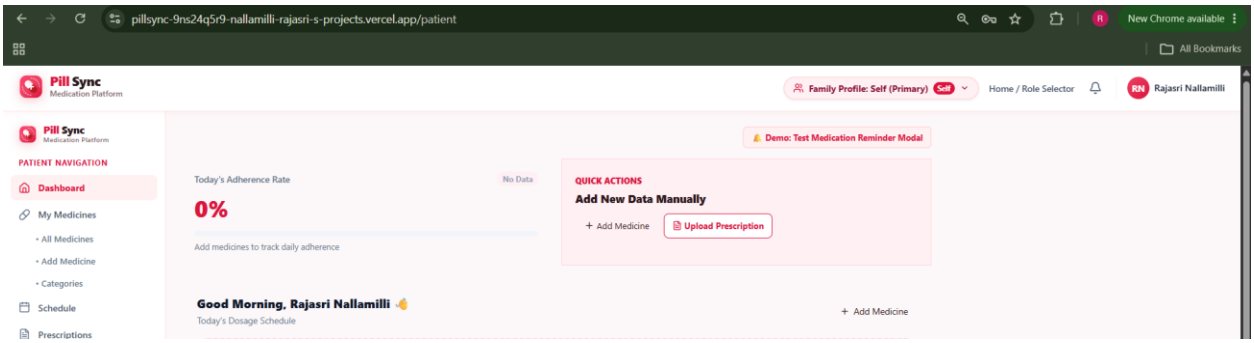
- Defining administrator permissions.
- Checking ownership or caregiver relationships.
- Restricting access to patient profiles according to the user's relationships.
- Determining accessible patient profiles using `User.accessible_patient_profiles()`.



4. Patient and Family Profile Management

Features represented in the code include:

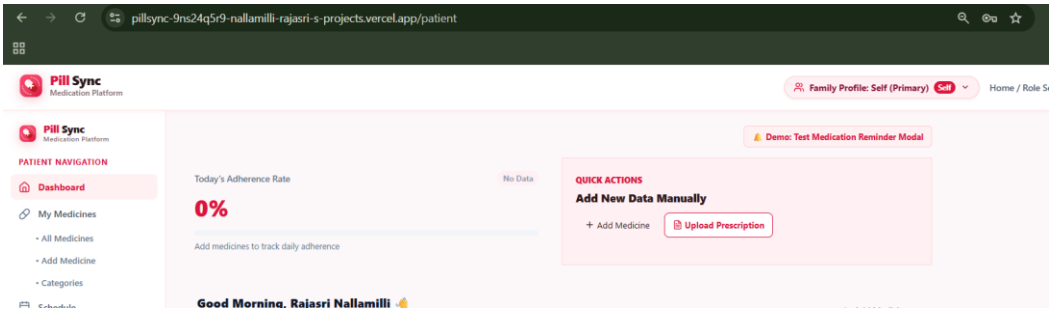
- Patient profile creation and management.
- Family profile selection.
- Multiple patient profiles.
- Primary patient-profile selection.
- Medical condition information.
- Emergency contact details.
- Caregiver-associated patient profiles.



5. Medicine Management

Functionalities represented in the project include:

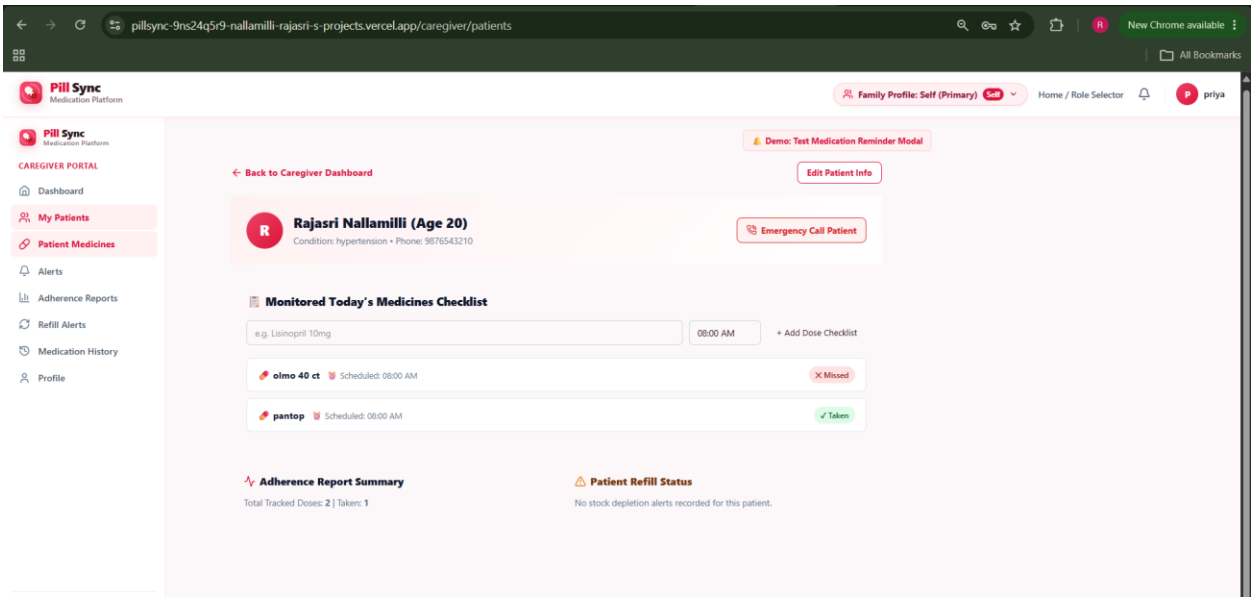
- Adding medicines.
- Viewing the medicine list.
- Maintaining dosage information.
- Recording medicine quantity and frequency.
- Categorizing medicines by health condition.
- Maintaining active medicine information.
- Managing medication schedules.
- Organizing medicines by patient profile.



6. Medication Reminders and Dose Actions

Implemented code structures include:

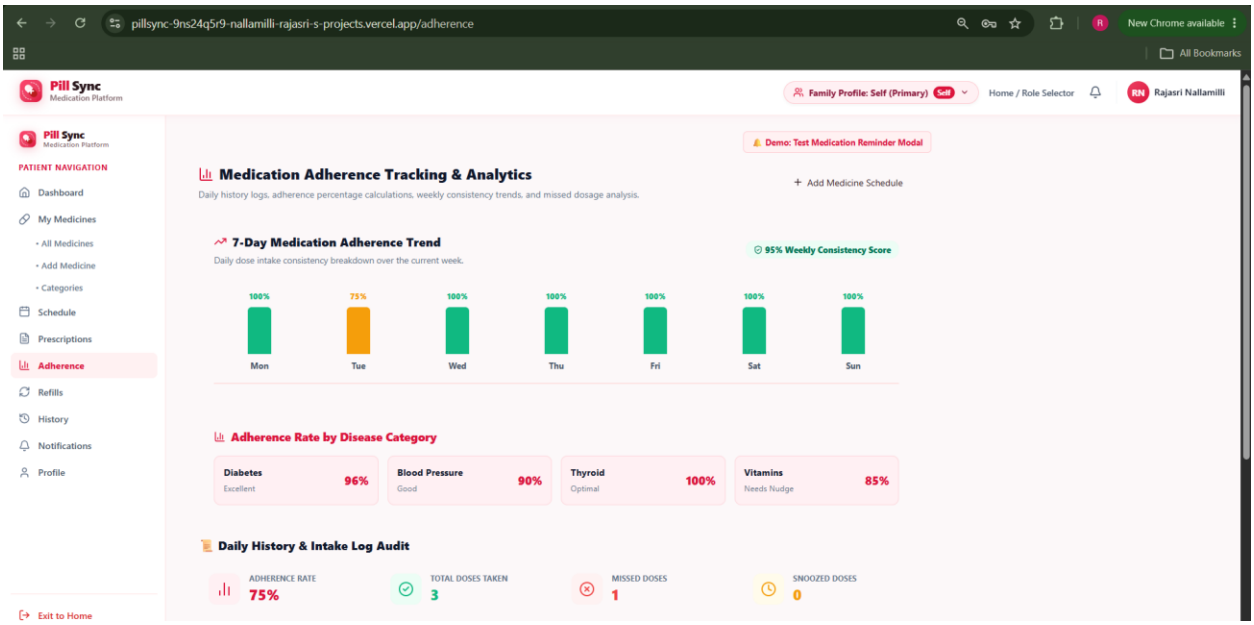
- Creating and maintaining reminder records.
- Recording a dose as taken.
- Marking a dose as missed.
- Snoozing a reminder.
- Skipping a reminder.
- Recording dose-related actions for adherence tracking.



7. Medication Adherence Tracking

Functionalities include:

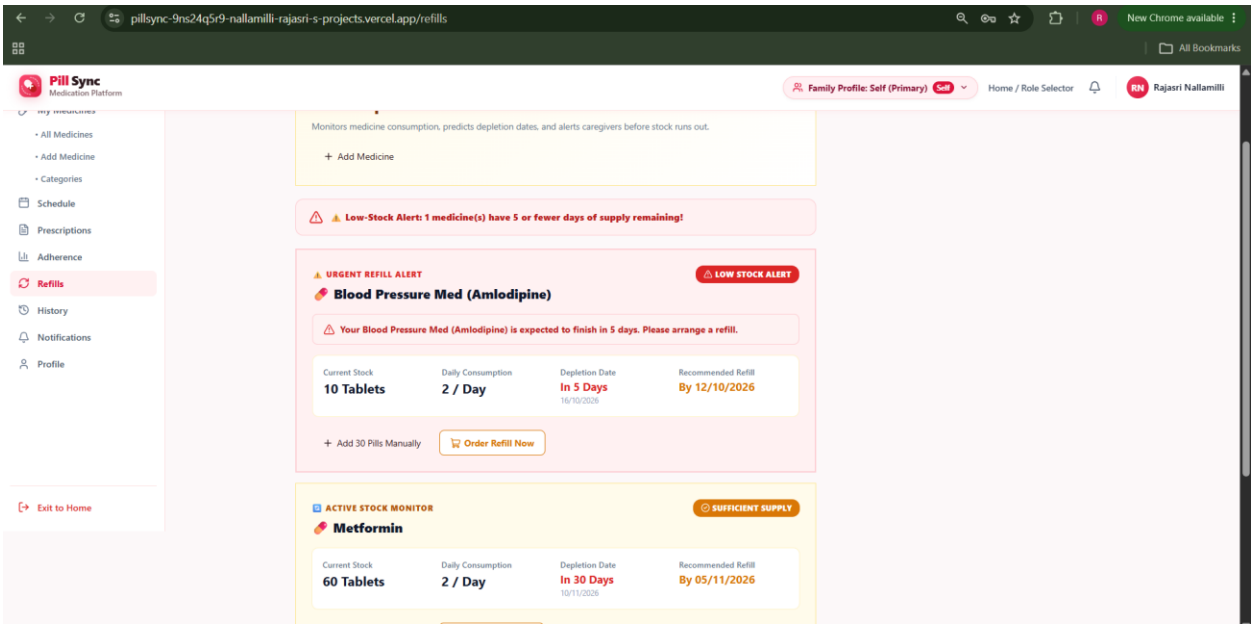
- Recording taken doses.
- Recording missed doses.
- Maintaining intake history.
- Tracking snoozed doses.
- Calculating the percentage of doses taken.
- Counting taken, missed, and snoozed records.
- Displaying medication history.



8. Refill Prediction

The engine accepts:

- Remaining medicine stock.
- Daily dosage frequency.
- Quantity consumed per dose.
- Missed-dose information.
- A configurable low-stock threshold.

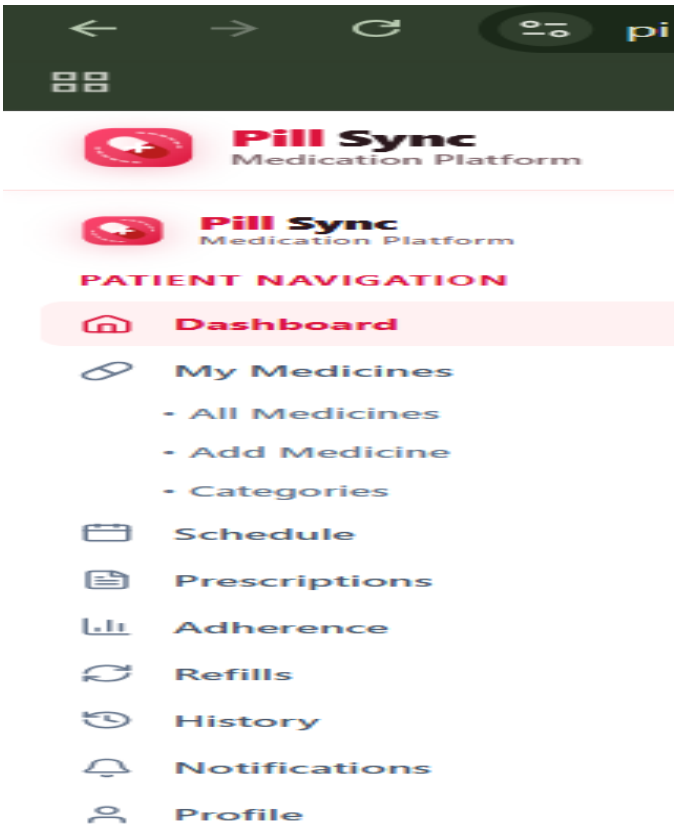


OTHER FEATURES INCLUDE :

- Prescription Management and OCR Workflow
- Notifications
- Administrator and Caregiver Portals
- Analytics & reporting

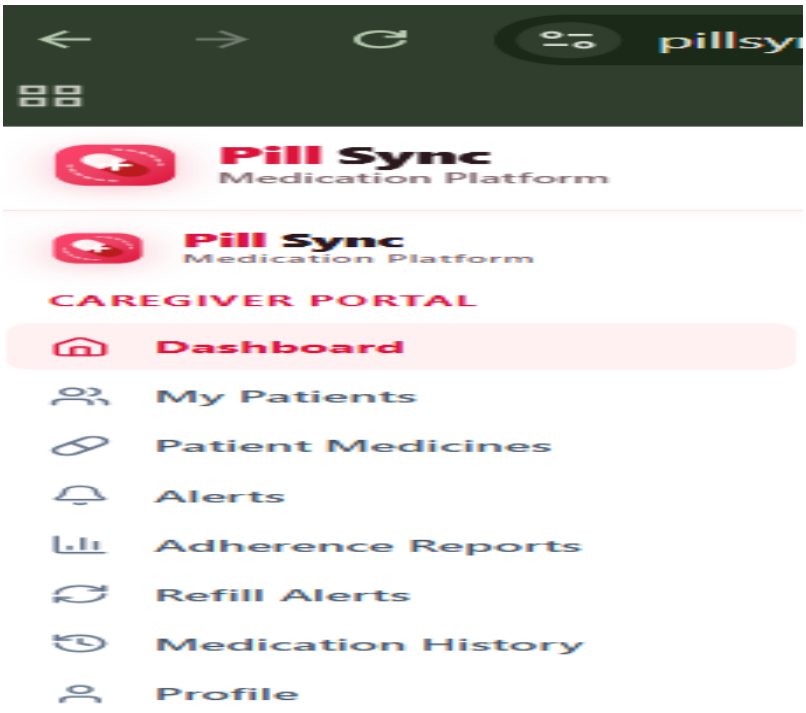
PATIENT FEATURES

- Patient registration and login.
- Patient profile management.
- Adding medicines.
- Viewing the medicine list.
- Medicine scheduling.
- Prescription management.
- Prescription upload and OCR workflow.
- Medication adherence tracking.
- Medication history.
- Refill prediction.
- Disease-based medicine categories.
- Notifications.
- Patient dashboard.



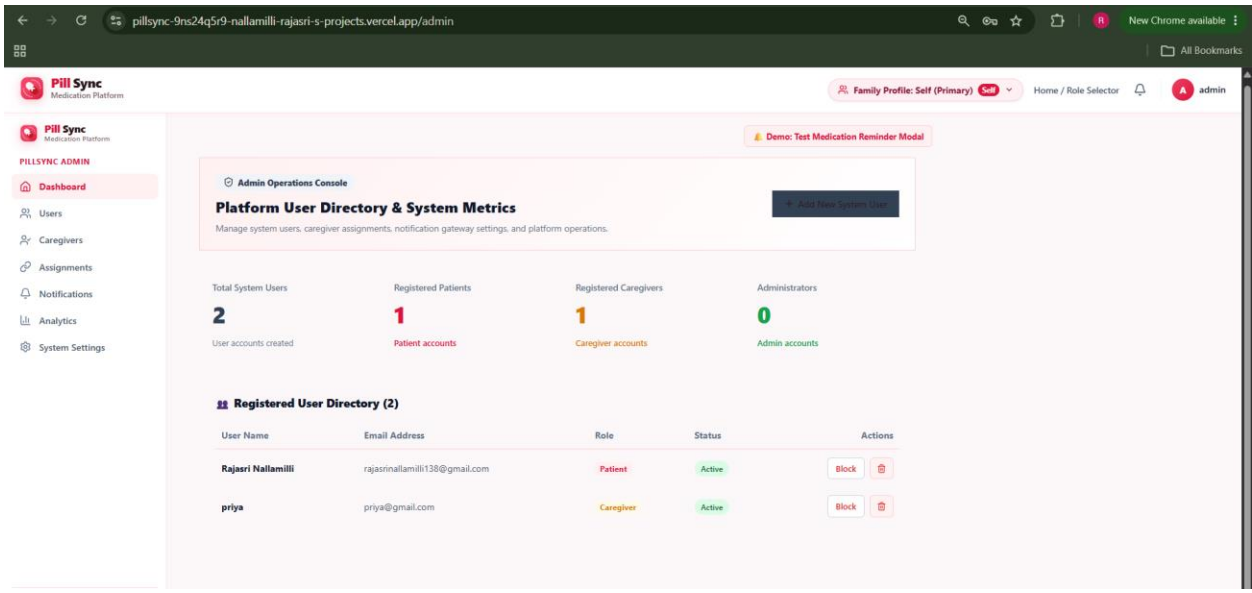
Caregiver Features

- Caregiver registration.
- Caregiver login.
- Caregiver dashboard.
- Patient detail pages.
- Family-profile support.
- Backend logic for determining accessible patient profiles.
- Role-specific permission structures.



Administrator Features

- Admin registration page.
- Admin login page.
- Admin dashboard.
- User management page.
- Notification-settings page.
- Administrator-specific permission classes.
- Backend user and authentication structures.


pillsync-9ns24q5r9-nallamilli-rajasri-s-projects.vercel.app/admin
New Chrome available
All Bookmarks
Pill Sync
Medication Platform
Family Profile: Self (Primary) Self
Home / Role Selector
admin
Pill Sync
Medication Platform
PILLSYNC ADMIN
Dashboard
Users
Caregivers
Assignments
Notifications
Analytics
System Settings
Demo: Test Medication Reminder Modal
Admin Operations Console
Platform User Directory & System Metrics
Manage system users, caregiver assignments, notification gateway settings, and platform operations.
Total System Users
2
User accounts created
Registered Patients
1
Patient accounts
Registered Caregivers
1
Caregiver accounts
Administrators
0
Admin accounts
Registered User Directory (2)
User Name
Email Address
Role
Status
Actions
Rajasri Nallamilli
rajasrinallamilli138@gmail.com
Patient
Active
Block
priya
priya@gmail.com
Caregiver
Active
Block